

Table A: Land-use DDABM components, assumptions, and data sources.

ABM Component	Sub-category	Element	Value / Assumption	Data Source
Agents	State	Land-use state per parcel	U (undeveloped), D (developed)	—
	Attributes	Demographics	Income, household composition, tenure status	Census microdata (static)
	Population	N (household agents)	Scaled to regional total; developer agents added	Census microdata (static)
	Behaviors	Move decision	Triggered by life-cycle events or dissatisfaction > threshold	Real-estate transactions (dynamic)
	Behaviors	Location choice	Utility-based bid (multinomial logit) over feasible parcels	Real-estate transactions (dynamic)
	Behaviors	Developer conversion	$U \rightarrow D$ if expected return > τ	Real-estate transactions (dynamic)
Interaction Rules	Bidding	Parcel competition	Highest bid wins; ties broken stochastically	Real-estate transactions (dynamic)
	Neighbor effects	Amenity update	Local density & land cover update aesthetic and price within Moore neighborhood	Historical land-use maps (dynamic)
	Agent–Environment	Feasibility	Zoning and growth-boundary layers constrain the feasible-parcel set	Zoning & policy layers (dynamic)
Environment	Grid	Spatial structure	Regular raster, 100 m × 100 m cells	Parcel cadastre (static)
	Layers	Zoning	Allowed uses and density caps per cell	Zoning & policy layers (dynamic)
	Layers	Accessibility	Network distance to CBD and transit	Road & transit GIS (static/dynamic)
	Layers	Amenity	Lakes, parks, views, neighborhood quality	GIS feature layers + hedonic regression (static)
Other Key Parameters				
Interventions	Policies	UGB, TOD upzoning, combined	Spatial constraint on feasible parcels; density-allowance multipliers near transit	Zoning & policy layers (dynamic)
Preference weights, τ , neighbor radius	Behavioral parameters	Parameters	Estimated via calibration	Real-estate + Historical land-use maps
Time step	Temporal resolution	Step size	1 year	—
Duration / Stop	Simulation length	End condition	40 years (1990–2030); stop at horizon or when developable land exhausted	—

Table B: Data-driven workflow summary for the land-use DDABM.

Stage	Details	Data / Input	Method	Output / Role
Data Inputs & Pipeline	Collection	Parcel cadastre, census microdata, NLCD/Landsat land-use maps, MLS transactions, zoning layers	County GIS export, USGS NLCD archive, census API, MLS/assessor feed	Raw datasets for initialization & calibration
	Pre-processing	All collected datasets	Parcel-to-raster rasterization (100 m); land-cover reclassification (U/D); spatial join of demographics to parcels	Cleaned 100 m grid with parcel attributes
	Analysis	Pre-processed data	Hedonic regression for amenity weights; k-means clustering of move patterns → preference classes; accessibility surface (network distance)	Initial amenity-weight estimates, preference-class assignments, accessibility layer
Model Evaluation	Calibration	Historical land-use maps (1990 → 2010 trajectory)	Pattern-oriented modeling (POM) with genetic algorithm over preference weights, developer threshold τ , neighbor-effect radius	Best-fit parameter set; multi-resolution Kappa > 0.70 over calibration period
	Output validation	Held-out 2010–2020 land-use maps	Multi-resolution Kappa; density-gradient and fractal-dimension comparison	Quantified spatial accuracy on unseen period
	Scenario validation	Post-policy implementations (e.g., adopted growth boundaries)	Simulated vs recorded post-policy land-use trends	Confirmed model responsiveness to policy levers
	Null-model contrast	Calibrated model vs random allocation and persistence null models	Pattern-metric comparison against null baselines	Demonstrates calibrated model clearly outperforms naive baselines
Scenarios (Simulation Experiments)	Baseline	Calibrated preferences; status-quo zoning	Standard land-use ABM run	Reference settlement trajectory
	Growth boundary	Spatial constraint around existing built-up area	Feasible-parcel set restricted to within UGB	Compacted development; potential leapfrog at boundary
	TOD upzoning	Density-allowance multiplier within 800 m of transit corridors	Modified zoning layer near transit	Polycentric structure; corridor densification
	Combined policies	UGB + TOD upzoning	Joint application of both constraints	Key scenario for DT-informed policy recommendation